

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL GCSE BIOLOGY

Paper 2

Tuesday 28 May 2019

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



Answer **all** questions in the spaces provided.

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0 1

This question is about inheritance.

0 1 . 1

Characteristics are determined by inherited genes and the environment.

Which characteristic is mainly determined by the environment?

[1 mark]

Tick (✓) **one** box.

Body mass

☐

Cystic fibrosis

☐

Eye colour

☐

0 1 . 2

Draw **one** line from each genetic term to its correct description.

[3 marks]

Genetic term

Description

Chromosome

structure in the nucleus that contains genes

DNA

a type of reproduction involving sex cells

Variation

the differences in characteristics between organisms of the same kind

having two different alleles for a characteristic

the chemical genes are made from



Fur colour in mice is an inherited characteristic.

The allele for black fur (B) is dominant to the allele for brown fur (b).

0 1 . 3

What is the genotype of a mouse with brown fur?

[1 mark]

Tick (✓) **one** box.

BB

☐

Bb

☐

bB

☐

bb

☐

Question 1 continues on the next page

Turn over ►



Two black mice were allowed to breed.

0 1 . 4

Complete **Figure 1** to show the possible genotypes of the offspring.

[2 marks]

Figure 1

		Male mouse	
		B	b
Female mouse	B	BB	
	b	Bb	

Key
 B Allele for black fur
 b Allele for brown fur

0 1 . 5

What is the ratio of black offspring to brown offspring in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

1 : 1

☐

1 : 4

☐

3 : 1

☐

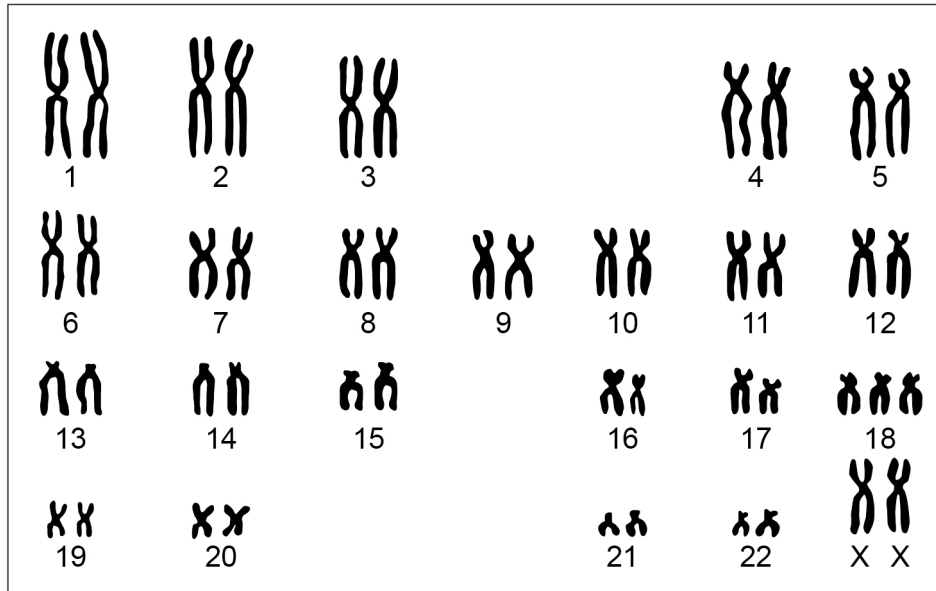
4 : 0

☐


Syndrome E is a disorder caused by having an abnormal number of chromosomes.

Figure 2 shows the chromosomes from one cell of a female with syndrome E.

Figure 2



0 1 . 6

Describe how the chromosomes in **Figure 2** differ from the chromosomes of a female who does **not** have syndrome E.

[1 mark]

0 1 . 7

Give the name of **one** other condition which is caused by an abnormal number of chromosomes.

[1 mark]

Question 1 continues on the next page

Turn over ►



Tests to find out whether babies have syndrome E can be carried out when the mother is 10 to 14 weeks pregnant. These tests do **not** harm the mother or baby.

Table 1 gives some information about two types of screening tests available.

These tests are carried out on the mother's blood.

Table 1

Test	What is tested	Accuracy of test (%)	The chance of getting a wrong positive result (%)
Test 1	Mother's hormone levels	85.0	2.0 – 3.0
Test 2	DNA from baby in mother's blood	97.0	0.3

0 1 . 8

Give **two** advantages of **Test 2** compared with **Test 1**.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____

0 1 . 9

A test on the fluid surrounding the baby (**Test 3**) can be carried out when the mother is 15 weeks pregnant.

Test 3 is 99% accurate at detecting syndrome E. However, it increases the risk of miscarriage.

Give **two** advantages of **Test 2** compared with **Test 3**.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____



Animal faeces (droppings) decay to form manure. Manure contains mineral ions that plants need for growth.

A gardener put the faeces from three different animals into separate containers. The containers had holes in their sides. After six weeks manure had formed in each container.

0 2 . 2 Why did the containers have holes in the sides?

[1 mark]

Tick (✓) **one** box.

To allow nitrogen to enter

☐

To allow oxygen to enter

☐

To allow the waste to cool down

☐

To allow the waste to dry out

☐

Table 2 shows the percentage of different mineral ions in each type of manure.

Table 2

Type of manure	Percentage of mineral ions (%)		
	Nitrate	Phosphate	Potassium
Cow	0.6	0.2	0.5
Horse	0.7	0.3	0.6
Rabbit	2.4	1.4	0.6

0 2 . 3 The gardener predicted it would be best to use rabbit manure to increase the growth of plants.

Suggest **one** reason that supports the gardener's prediction.

[1 mark]



The gardener set up an investigation to test his prediction.

This is the method used.

1. Put some garden soil in four pots.
2. Add 10 bean seeds to each pot.
3. Cover the seeds in the first pot with garden soil.
4. Cover the seeds in each of the other three pots with a different type of manure.
5. Keep all four pots in a warm and light place.
6. Water the pots regularly.
7. After 6 weeks measure the height of each bean plant.
8. Calculate a mean height for the bean plants in each pot.

0 2 . 4

What is the independent variable in the investigation?

[1 mark]

0 2 . 5

Why did the gardener grow some bean plants in a pot with no manure?

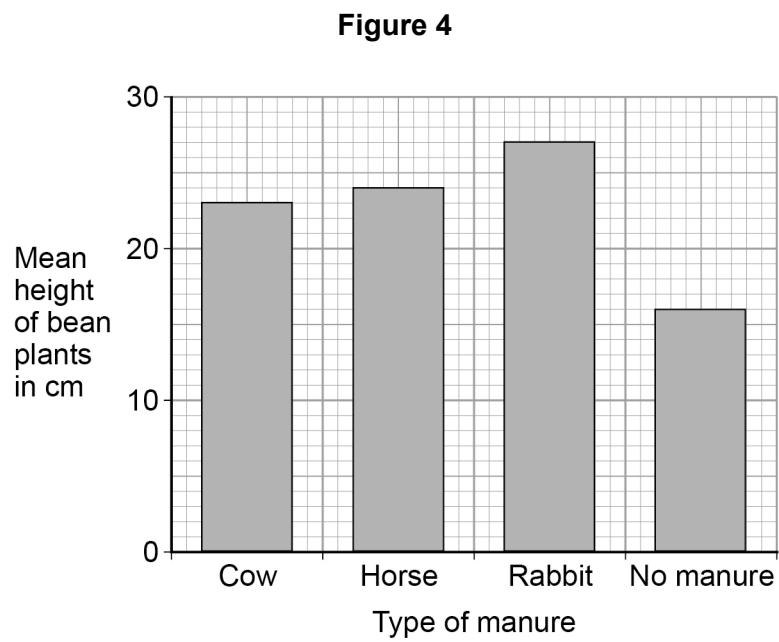
[1 mark]

Question 2 continues on the next page

Turn over ►



Figure 4 shows the gardener's results.



0 2 . 6

Describe the effect of using different types of manure on the growth of the plants.

[3 marks]



The farmer grows wheat in the same field every year. The wheat is harvested and sold. If manure is **not** added to the soil the mass of wheat harvested each year will decrease.

[4 marks]

[illegible]

17

Turn over for the next question

Turn over ►



0 3

This question is about blood.

0 3 . 1

Give the name of the liquid part of the blood.

[1 mark]

0 3 . 2

The liquid part of the blood transports red blood cells, white blood cells, platelets and dissolved substances around the body.

Name **two** dissolved substances transported by the liquid part of the blood.**[2 marks]**1

2

0 3 . 3Red blood cells do **not** have a nucleus.

Suggest how this adaptation helps the red blood cells to carry out their function.

[2 marks]



0	3	.	4
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Platelets release chemicals which begin the blood clotting process.

Describe the sequence of events that leads to a blood clot forming.

[3 marks]

Question 3 continues on the next page

Turn over ►



Table 3 shows the distribution of the human blood groups **A**, **B**, **AB** and **O** in five countries.

Table 3

Country	Percentage of the population with each blood group (%)			
	A	B	AB	O
Chile	8.9	3.4	1.0	86.7
India	21.4	40.3	9.4	28.9
Japan	40.2	19.8	9.9	30.1
Mexico	31.4	8.4	1.7	58.5
United Kingdom	42.0	10.5	3.4	44.1

0 3 . 5

Calculate the difference in the mean percentage with blood group **O** compared to the mean percentage with blood group **AB**.

[2 marks]

Difference = _____ %

0 3 . 6

Compare the distribution of blood group **A** with blood group **B** between the five countries.

[3 marks]



Question 3 continues on the next page

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A person's blood group is determined by which antigens (**A** or **B**) are present on the surface of the red blood cells.

Table 4 shows the antigens present in the different blood groups and the antibodies which are found in the blood.

Table 4

Blood group	Antigen on red blood cells	Antibody found in the blood
A	A	anti-b
B	B	anti-a
AB	A and B	none
O	none	anti-a and anti-b

A blood transfusion involves transferring blood from one person (donor) into another (the recipient).

It is essential to use the correct blood group to make sure the transfusion is safe.

If the incorrect blood group is used, the antibodies of the recipient will attack the antigens on the red blood cells from the donor.

0 3 . 7

Complete **Table 5** to show which combinations of blood groups would be safe (✓) in a transfusion and which would be unsafe (✗).

[3 marks]

Table 5

Blood group of donor	Blood group of recipient			
	A	B	AB	O
A				
B				
AB				
O	✓	✓	✓	✓

Key

✓ = safe transfusion

✗ = unsafe transfusion



0 3 . 8Explain why blood group **O** can be safely donated to recipients of any blood group.**[3 marks]**

19**Turn over for the next question****Turn over ►**

0 4

Students set up an investigation to determine the concentration of an unknown sugar solution.

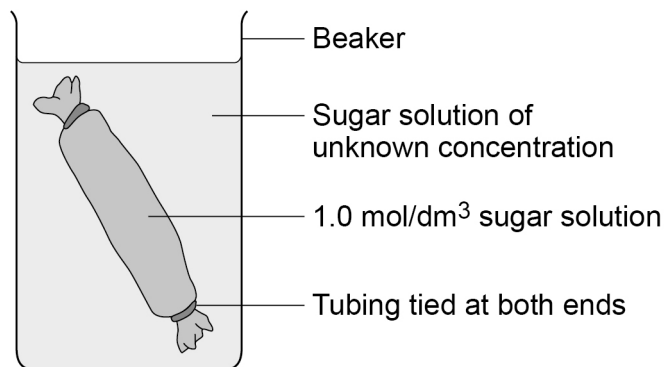
The students used tubing made from a partially permeable membrane.

This is the method used.

1. Cut a 10 cm length of tubing and tie one end.
2. Fill the tubing with 10 cm³ of 1.0 mol/dm³ sugar solution and tie the other end.
3. Measure and record the mass of the tubing.
4. Repeat steps 1 to 3 with five other known concentrations of sugar solution.
5. Place each length of tubing into a separate beaker containing sugar solution of the same unknown concentration.
6. Leave the beakers for 12 hours at 20 °C.
7. Remove each length of tubing from the beakers.
8. Measure and record the mass of each tubing.

Figure 5 shows the apparatus set up for the 1.0 mol/dm³ sugar solution.

Figure 5



0 4 . 1

Give **two** variables that were controlled in the investigation.

[2 marks]

- 1 _____
- _____
- 2 _____
- _____



0	4	.	2
---	---	---	---

A teacher said that the students should roll each length of tubing on a paper towel before weighing it.

Suggest why.

[1 mark]

Question 4 continues on the next page

Turn over ►



Table 6 shows the students' results.

Table 6

Concentration of sugar solution in tubing in mol/dm ³	Mass of tubing at start in g	Mass of tubing after 12 hours in g	Change in mass in g	Percentage change in mass (%)
0.0	10.24	8.15	−2.09	−20
0.2	10.26	8.41	−1.85	
0.4	10.59	9.11	−1.48	−14
0.6	10.15	9.22	−0.93	−9.0
0.8	10.46	10.77	+0.31	+3.0
1.0	10.30	12.57	+2.27	+22

0 4 . 3

Calculate the percentage change in mass for the tubing containing 0.2 mol/dm³ sugar solution.

Give your answer to two significant figures.

[3 marks]

Percentage change = _____ %



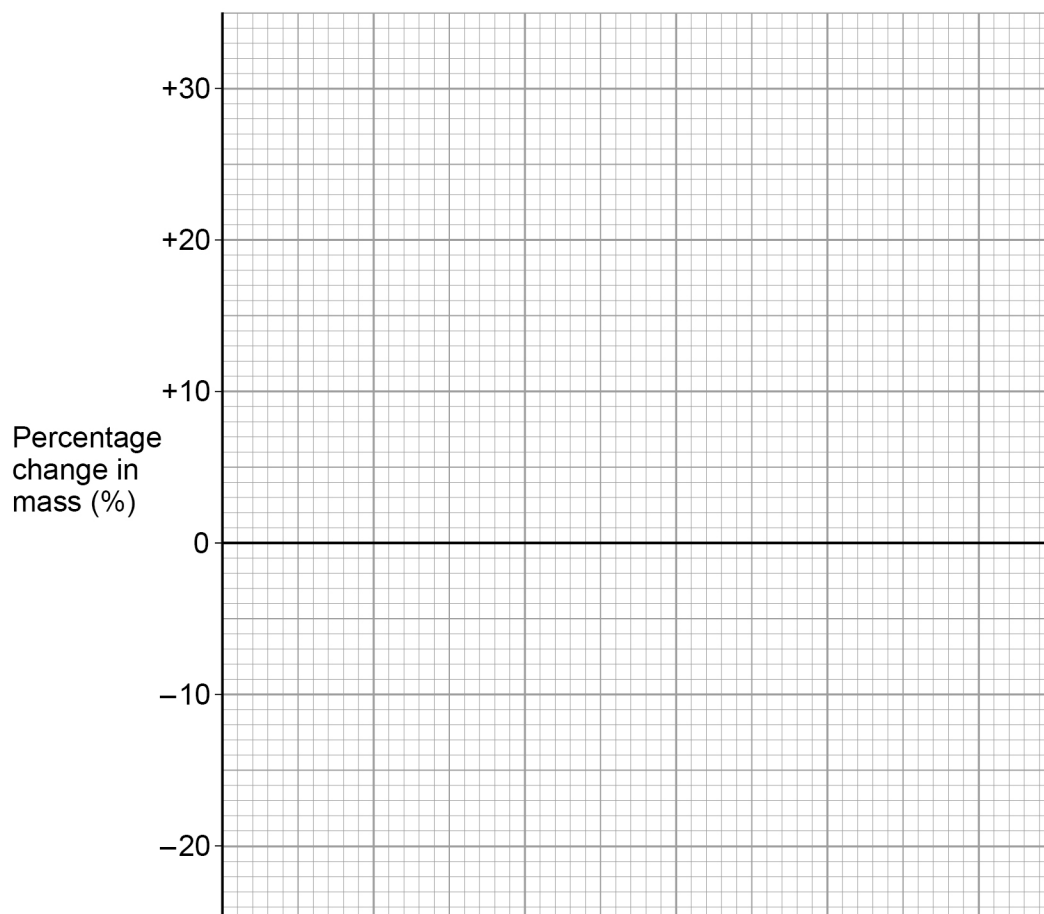
0 4 . 4 Plot the percentage change in mass from **Table 6** on **Figure 6**.

You should:

- label the x-axis
- use a suitable scale
- draw a line of best fit.

[4 marks]

Figure 6



0 4 . 5 Determine the concentration of the unknown sugar solution in the beakers.

Use your graph in **Figure 6**.

[1 mark]

Concentration = _____ mol/dm³

Question 4 continues on the next page

Turn over ►



0	4	.	6
---	---	---	---

The answer you gave in question **04.5** may **not** be an accurate value for the concentration of the sugar solution in the beakers.

How could the students improve the method to get a more accurate value?

[2 marks]

0	4	.	7
---	---	---	---

Suggest what the students could do to correct a zero error on the balance.

[1 mark]

0	4	.	8
---	---	---	---

Explain how a zero error on the balance would cause the students' results to differ from the true values.

[2 marks]



0	4	.	9
---	---	---	---

Storing fruit in a concentrated sugar solution prevents the fruit decaying.

Explain how a concentrated sugar solution prevents fruit decaying.

[4 marks]

20

Turn over for the next question

Turn over ►



0 5

This question is about evolution.

Some types of crab have claws of different sizes.

The male fiddler crab has one enlarged claw. Bigger claws are stronger.

The large claw is used for attracting females and to fight other males.

Figure 7 shows a male fiddler crab.

Figure 7



Male fiddler crab

0 5**. 1**

Use Lamarck's theory to explain how the large claw of the male fiddler crab evolved.

[3 marks]



Question 5 continues on the next page

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A lot of mining took place in Europe over 200 years ago. The mining caused soil to become contaminated with waste.

Figure 8 shows two different populations of one species of grass growing near a mine.

Figure 8

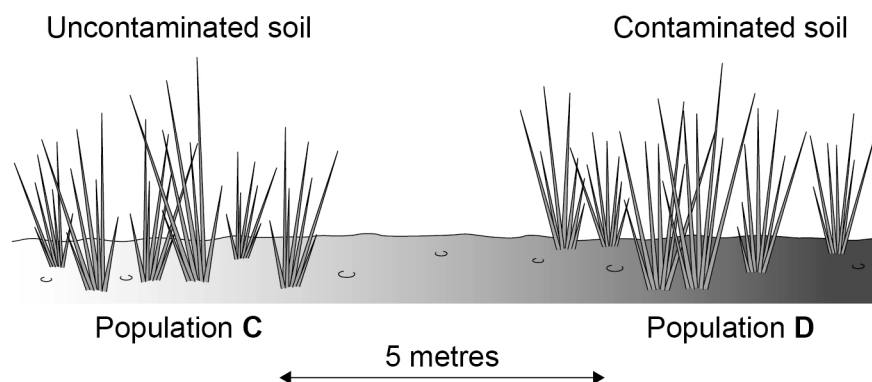


Table 7 gives some information about the two populations of grass.

Table 7

	Population C	Population D
Percentage of plants able to survive in the contaminated soil (%)	10	95
Month of flowering	June	May

Describe how the two populations of grass could evolve to become two separate species.

[illegible]

9

0 6

This question is about animal behaviour.

0 6

1

Animals use a variety of chemical, sound and visual signals to communicate.

Figure 9 shows a bird with a red throat sac.

The bird can inflate the throat sac by swallowing air.

Figure 9

Red
inflated
throat
sac



Black
feathers
of body

Suggest how the bird uses the throat sac to communicate.

[3 marks]

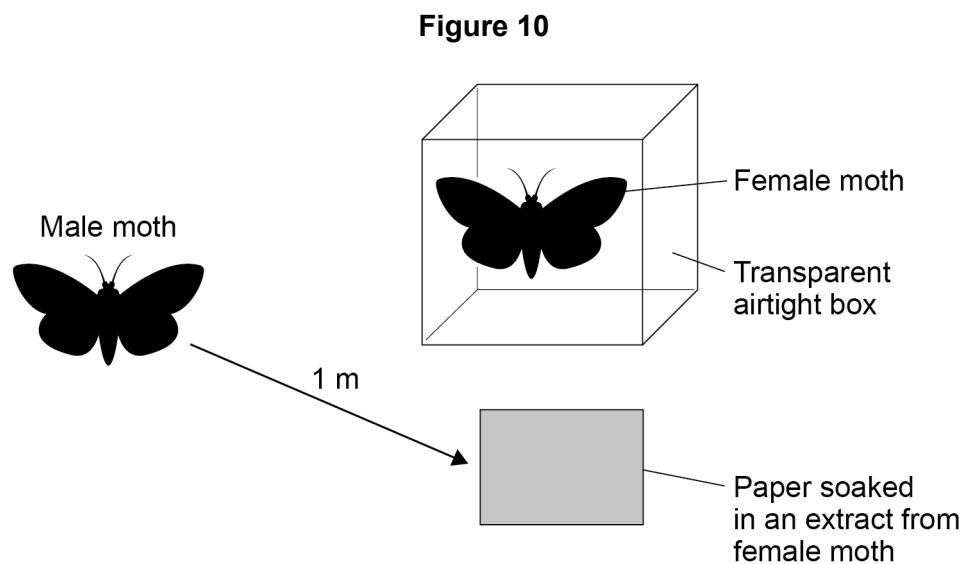


A scientist investigated communication in moths.

Female moths were kept in a transparent airtight box. Paper was soaked in an extract from female moths. Male moths were released 1 m away from both the transparent airtight box and the paper.

The scientists recorded the direction of flight of the male moths.

Figure 10 shows how the investigation was set up.



0 6 . 2

Suggest why the male moths flew to the paper and **not** to the female moths in the box.

[3 marks]

Turn over ►



0 6 . 3 **Figure 11** gives some information about whales.

Figure 11

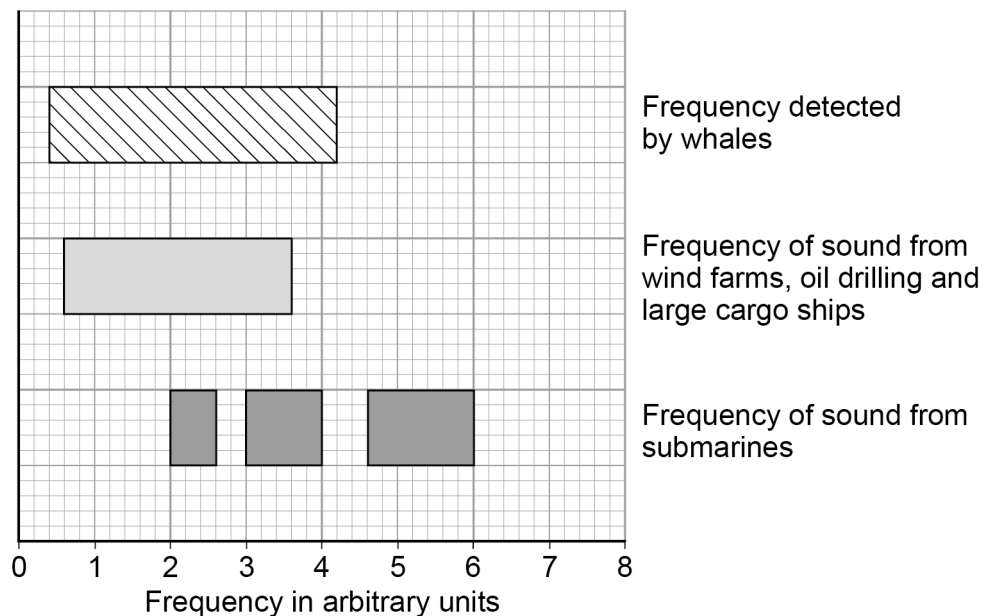
Sound pollution from human activities in the ocean is linked to the death of whales.

Whales communicate using sound. Whales produce sound waves which bounce off objects in the water and return to the whales as an echo.

The whales can measure distance using the echoes they detect.

Figure 12 shows the frequency range detected by whales and the frequency ranges of sounds produced from some human activities.

Figure 12



Evaluate the claim that sounds from human activities caused the death of the whales. Use information from **Figure 11** and **Figure 12** in your answer.

[illegible]

11

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